Threshold Gating and Information Disclosure in Observational Learning

Abstract

Early public actions can amplify noisy signals and generate inefficient information cascades in sequential financial decisions. We test whether threshold gating improves collective accuracy across disclosure regimes. In an agent-based model, ten agents receive binary private signals with 0.60 accuracy and decide sequentially. Across 3,000 independent groups per treatment, we compare pure action observation (PAO), optional disclosure (OD), selective disclosure (SD), and incentivized optional disclosure (IOD). A dual-threshold mechanism delays weak public signals and gives silence directional meaning only after public evidence becomes sufficiently strong. An LLM-based module adds heterogeneous information-processing weights for a small-sample consistency check. Without threshold gating, screened or incentivized disclosure performs best: mean accuracy over the last five decisions is 0.888 under SD and 0.856 under IOD, compared with 0.753 under PAO and 0.660 under OD. The baseline thresholds raise accuracy from 0.753 to 0.798 under PAO and from 0.660 to 0.697 under OD by limiting early reliance on noisy public histories. Effects are heterogeneous when disclosure already filters information: accuracy rises from 0.888 to 0.910 under SD but falls from 0.856 to 0.843 under IOD, revealing a cost from delaying useful signals. Results remain directionally stable across random seeds. Sensitivity tests identify effective ranges of 2-3 for the public-information threshold and 2-4 for the silence threshold. LLM augmentation increases behavioral variation while preserving the broad advantage of screened or incentivized disclosure. Threshold gating is therefore a targeted corrective for weak information environments, not a universal addition to disclosure rules. Effective design requires matching information-use thresholds to the screening capacity of the underlying disclosure regime.

Keywords: Observational learning; Information disclosure; Threshold gating; Information cascades; Agent-based simulation; Large language models